

General Anesthesia and Severe Maternal Morbidity Among Nulliparous Women: nuMoM2b Cohort

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The LABOR Study
Leveraging Anesthesia-Based Obstetric
Risk for Maternal Safety

Background

Severe maternal morbidity (SMM) affects approximately 50,000 U.S. deliveries annually and remains a leading driver of preventable maternal harm. Anesthesia care is closely intertwined with outcomes, yet the relationship between anesthetic modality and SMM is understudied.

The Exposure Challenge General anesthesia (GA) occurs in only 0.7% of this cohort — a rare exposure creating extreme propensity scores and limited covariate overlap. GA selection is strongly confounded by clinical urgency: emergent conditions drive both GA selection and SMM risk independently.

Our Analytic Solution Overlap weighting (OW) targets inference where covariate overlap is strongest, downweighting extreme propensity scores without discarding participants.

Objective

Does GA carry higher SMM risk than neuraxial anesthesia among nulliparous individuals, and which propensity score method performs best under rare-exposure conditions?



Sample Characteristics

Characteristic	GA (n=54)	Neuraxial (n=7,744)
Mean age, years	26.2	27.0
Mean BMI, kg/m ²	28.0	26.5
Mean gest. age, wks	34.1	39.3
Non-Hispanic White	53.7%	61.3%
Non-Hispanic Black	18.5%	13.0%
Hispanic	20.4%	17.2%
Cesarean delivery	88.9%	29.2%
SMM (crude)	31.5%	9.9%

Largest pre-adjustment imbalances: gestational age (SMD 2.23), cord prolapse (SMD 1.77), cesarean delivery (SMD 1.20). After overlap weighting (OW), mean |SMD| decreased from 0.209 to 0.137.

Methods

Cohort nuMoM2b: multicenter prospective, nulliparous pregnancies (2010–2013). Analytic sample: n=7,798 (GA: 54 [0.7%]; neuraxial: 7,744 [99.3%]).

Outcome CDC composite SMM index — 21 indicators including transfusion, invasive ventilation, DIC, hysterectomy, and sepsis, etc.

Covariates Age, BMI, gestational age, race/ethnicity, education, insurance (complete-case; <2% missing per variable).

Primary Overlap weighting (OW), Optuna-tuned logistic regression propensity scoring; Average treatment effect in the overlap population (ATO) estimand.

Sensitivity 1:1 propensity score matching (PSM) with logistic regression, random forest, and XGBoost.

Post-hoc Cesarean indication profiles to characterize residual urgency confounding.

Results

54 of 7,798
Rare general anesthesia

GA: 54 (0.7%) | Neuraxial: 7,744

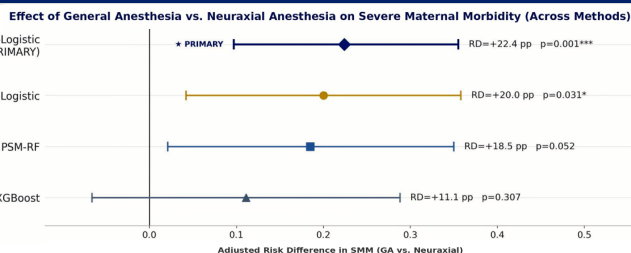
3x HIGHER
Crude SMM rate in
GA vs. neuraxial

31.5% vs. 9.9% (unadjusted)

+22.4 percentage points
Higher SMM risk in GA vs. neuraxial

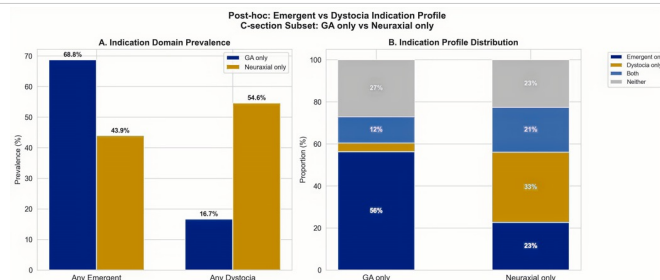
95% CI: 9.7 to 35.5, p<0.001 | Overlap weighted

Primary & Sensitivity Analyses — All Directionally Consistent



Overlap weighting (OW)-Logistic is the primary analysis. PSM analyses are sensitivity checks using three different matching algorithms. All estimates point in the same direction. OW produced the most stable estimate for this rare-exposure setting. XGBoost attenuation is a known technical limitation when one group is very small; not a contradictory finding.

Post-Hoc: Residual Urgency Confounding — GA Disproportionately Emergent



GA-only deliveries were more often emergent (68.8% vs. 43.9%); neuraxial-only more often dystocia-related (54.6% vs. 16.7%). After overlap weighting, urgency-related standardized mean differences (SMD) remained poor (|SMD|>0.20) across all indication variables, confirming residual confounding by indication that statistical adjustment cannot fully resolve.

Key Limitations

- **Confounding by Indication** GA is selected in the most urgent, highest-risk deliveries. Emergent cesarean, hemorrhage, and failed neuraxial block independently cause SMM. Statistical adjustment cannot fully separate anesthetic effect from delivery complexity.
- **Residual Urgency Imbalance** After overlap weighting, urgency standardized mean differences (SMD) remained poor (|SMD|>0.20): unscheduled/urgent Cesarean section (+0.39), arrest of dilation (−0.26), cord prolapse (+0.36). The observed association reflects an adjusted descriptive estimate — not a causal effect.
- **Small GA Group (n=54, 0.7%)** Reduces power and precision; wide confidence intervals reflect this constraint. Chi-square assumptions are borderline at this cell size.
- **Generalizability** Nulliparous only, enrolled 2010–2013, U.S. based. Findings may not reflect contemporary practice in global or multiparous populations.

Conclusions

- ✓ **GA as a Marker, Not Simply a Cause** The +22.4 percentage point adjusted excess severe maternal morbidity most likely reflects a combination of underlying patient risk, delivery urgency, and general anesthesia's physiological sequelae, which are components that cannot be cleanly separated.
- ✓ **Consistent Signal Across Methods** Direction consistent across overlap weighting (OW), propensity score matching (PSM)-logistic, PSM-random forest, all neuraxial subtypes, and all subgroups. OW produced the most stable estimates under rare-exposure conditions.
- ✓ **Methodological Contribution** Overlap weighting (OW) outperformed matching-based approaches. Post-hoc confounding characterization is as critical as estimator selection in rare-exposure obstetric research.
- ✓ **Clinical Implication** General anesthesia (GA) patients warrant recognition as high-risk and enhanced perioperative vigilance — regardless of whether GA caused or simply marked that risk.

Analyses conducted in Python 3.14 and R 4.5. Hyperparameter optimization via Optuna. Emory IRB ID: STUDY00009239.

Funding

This research has been supported in part by the **American Association of Nurse Anesthesiology (AANA) Foundation** (2025-G-8).

Partially funded through the **National Institutes of Health National Institute of Nursing Research** grant number T32NR020778 (PIs: D. Bruner and V. Hertzberg) for trainee Sheena Warner. The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health.



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